

# Ricardo (Zhicong) Xie

MECHANICAL ENGINEERING | PROTOTYPING, TESTING AND SIMULATION

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## PROFILE

- Mechanical engineering master's student who builds real hardware: designed, fabricated and wired a wearable energy-harvesting backpack prototype end to end, and uses CFD (Fluent) and FEA (Abaqus) to check designs.
- **Available full-time late November 2026 – late February 2027** (University of Auckland summer break).

## EDUCATION

### University of Auckland | Master of Engineering Studies (Mechanical)

Feb 2026 – Jul 2027 (expected)

GPA 6.0/9.0 | Research Methods (A), Advanced Computational Fluid Dynamics (A-), Composite Materials (B)

### Huazhong Agricultural University | BEng Mechanical Engineering (Mechatronics)

2021 – 2025

Mechanical design, electrical and electronic engineering, sensors, embedded control, mechatronic integration

## PROJECTS

### Energy-Harvesting Backpack Prototype | Lihua Tang Group, University of Auckland

Jul 2026 – present

- **Design:** modelled and designed the suspended-load platform in SolidWorks: twin guide rods, parallel elastic bands and a rack-and-pinion drive that turns  $\pm 35$  mm of payload travel into generator rotation (70 mm stroke  $\approx$  1 rev), designed for payloads of tens of kg. A fixed hole pattern lets the drive be upgraded without recutting the plates.
- **Build:** laser-cut the acrylic plates, 3D-printed parts, welded the generator support structure and assembled the platform; found a printed-hole/M5 clearance problem, corrected the tolerance and updated the CAD.
- **Wiring and fault-finding:** wired every coil of the group's generator prototype to a single output. When it gave no output, checked continuity joint by joint with a multimeter and traced the fault to a **coil lead worn through by friction** on an internal contact surface; repaired it and restored output.
- **Testing:** took part in doctoral-researcher-led shaker and oscilloscope tests (**volt-level open-circuit voltage** at the coil terminals) and demonstrations of walking-powered phone charging and a pulse oximeter.

### Solar Chimney Natural Convection and Conjugate Heat Transfer | ANSYS Fluent / CFX

Course research project

- Built a 2D axisymmetric three-region conjugate heat transfer model of buoyancy-driven flow; mesh independence across three grids (36,847-cell production mesh, 0.90% outlet-velocity difference to the fine mesh).
- Found that the reference paper's 12.2 m/s draught would need a  $\sim 299$  K temperature rise, inconsistent with its reported 42–46 °C, and treated it as an uncertain benchmark in the technical report.

### Composite Formula SAE Front Wing Structural Analysis | Abaqus

Course project

- S4R shell and composite layup model under aerodynamic, lateral and 4.1 g bump cases; identified the governing case with  $\sim 74\%$  peak stress utilisation.

### Dynamic Livestock Weighing System | STM32, HX711

2024 · Undergraduate

- Developed load-cell acquisition, embedded logic and serial/Bluetooth transfer for weighing moving animals; registered as Software Copyright No. 2024SR0141888.

## SKILLS

- **CAD / CAE:** SolidWorks (proficient: parts, assemblies, drawings); ANSYS Fluent/CFX, Abaqus, AutoCAD/DXF (working); MATLAB/Simulink (basic)
- **Programming:** Python (real-time device-control IoT scripts in a UoA postgraduate course, data processing, small automation tools); embedded C on STM32 (basic)
- **Hands-on:** 3D printing, laser cutting, CNC training, arc welding and electronics soldering, multimeter fault-finding, mechanical assembly and bench testing
- **Languages:** English (IELTS 7.0; CET-6), Mandarin (native)

## LEADERSHIP AND COMMUNITY

- **Photography Director**, Chinese Postgraduate Society (CNPg), New Zealand (2026 – present)
- **Head of Student News Department**, Huazhong Agricultural University: led news writing and photography, task allocation and deadlines
- **Volunteer**, Auckland Library of Tools (2026 – present); AUSA orientation volunteer